

At what size each instrument begins to see: Haghia Triada, Susa and Uruk measured by distribution

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Abstract

Three early administrative archives written in scripts whose languages are not understood, or understood only in part, are measured with a single set of distributional instruments: the Linear A tablets and roundels of Haghia Triada (Crete, c. 1450 BC; 1,117 sign-group tokens, 224 documents with two or more sign-groups), the Proto-Elamite tablets of Susa (Iran, c. 3100-2900 BC; 8,282 entries, 1,594 tablets) and the proto-cuneiform tablets of Uruk IV-III (Mesopotamia, c. 3350-3000 BC; 23,272 entries with numerals, 6,387 tablets). The same quantities are computed in each: the anatomy of the entry, the rate of singletons by length of sign-string, the behaviour that separates counted things from parties (recurrence within a document with different quantities), headings and totals, the exclusivity of vocabulary by document type, co-occurrence and co-exclusion against a fixed-margin null, the numeral systems by object class, and, where hands are attributed, the share of vocabulary explained by the scribe. The three archives share an anatomy (heading, entries of sign-string plus quantity, totals) and a partition of their sign-groups into a closed class of short, repeated commodity terms and an open class of long, singleton names; they differ in what each corpus can show at its size. Co-exclusion between document types, invisible at Haghia Triada, is measurable at Susa (13 pairs), and the inheritance of numeral systems from Uruk to Susa can be tested by structure (sexagesimal inherited, capacity adapted, decimal invented, bisexagesimal not taken). At Haghia Triada, uniquely, the attribution of hands allows the vocabulary to be assigned to scribes rather than to document types. The comparison places Minoan bookkeeping within the typology of early Near Eastern administration without reading a word, and gives, for each instrument, the corpus size at which it begins to see.

Keywords: Linear A, Proto-Elamite, proto-cuneiform, administrative archives, null models, co-exclusion, numeral systems, scribal hands.

1. Why compare three archives that cannot be read

The three corpora share the condition that makes them difficult and the one that makes them comparable. None has a bilingual: Linear A is not read; Proto-Elamite is not read and its language is unknown; proto-cuneiform is half read, through the continuity of its logograms and numerals into Sumerian cuneiform (Nissen, Damerow and Englund 1993; Englund 1998). All three are, on the other hand, accounting archives of a palace or institution, with the same document type: the tablet that records entries of a quantity of something associated with someone, with a heading and sometimes a total. The literature of each has described that anatomy separately (Schoep 2002 and Montecchi 2010 for Haghia Triada; Dahl 2005, 2019 and Damerow and Englund 1989 for Susa; Englund 1998 for Uruk), but it has not been measured

with the same instrument in all three, and so it is not known which features belong to early bookkeeping in general and which to each tradition.

The comparison also has a methodological value. The apparatus used here (Acedo 2026c) produces on Linear A a series of negative results that were attributed to the size of the corpus: no measurable co-exclusion, no reliability of network descriptors, no power to separate document types. Applied to sister corpora five and twenty times larger, it allows one to say for each instrument at what size it begins to see, and thus to turn those negatives into a power curve.

1 bis. The three archives

Haghia Triada, c. 1450 BC. A villa of the Mesara plain in Neopalatial Crete; its archive, some 150 tablets and 150 nodules and roundels, survives because the building burned in the Late Minoan IB destructions, and records the last year of an administration that allocated grain, oil, wine, figs, livestock and aromatics, with twenty-one scribes working at the same time (Schoep 2002; Montecchi 2010). There is no temporal stratification: it is a still photograph.

Susa, c. 3100-2900 BC. Capital of the Khuzestan plain, in contact with Uruk and at the same time independent of it; its 1,500 Proto-Elamite tablets are accounts of grain, livestock, personnel and derived products, with numeral systems of its own and a recurrent institutional header, and they are nearly all that survives of that administration; the periphery (Malyan, Tepe Yahya, Sialk, Sofalin) adds some fifty more (Damerow and Englund 1989; Dahl 2005, 2019).

Uruk IV-III, c. 3350-3000 BC. The earliest known written administration, in Lower Mesopotamia; some 5,000 tablets from the city and its surroundings, with the numeral systems from which those of Susa derive, and with a repertoire of logograms whose meaning is partly known from their continuity into Sumerian cuneiform (Nissen, Damerow and Englund 1993; Englund 1998). It is the only one of the three whose content is half understood.

2. Data and instruments

Linear A: GORILA (Godart and Olivier 1976-1985) in the form of the LinearA Explorer (Hogan 2022), with the hands of GORILA V and Younger's commentaries (2024); verification in SigLA (Salgarella and Castellan 2020). Proto-Elamite and proto-cuneiform: the transliterations and catalogue of CDLI (export of September 2026), with the numeral systems according to Damerow and Englund (1989) and Dahl (2019). In all three, the unit is the entry (a numbered line with a sign-string and a quantity); the sign-string of Linear A is the syllabic sign-group, that of Proto-Elamite and proto-cuneiform the sequence of non-numerical signs of the entry.

Instruments, in the same order in the three corpora: (1) size and singleton rate of the strings; (2) singletons by string length; (3) recurrence within a document with different quantities by length class; (4) headings (first entry) and totals; (5) vocabulary monopolies by document type or by numeral system; (6) co-occurrence and co-exclusion with the curveball null (Strona et al. 2014) on the documents $\times$ frequent strings matrix; (7) numeral systems by object class; (8) hands, where they exist. Nulls and thresholds are those described in Acedo (2026c): permutation preserving margins or lengths, $p < 0.01$ with one thousand permutations for new claims.

3. Results

3.1 The anatomy is the same

The three archives write the entry as sign-string plus quantity, open with a heading (SA-RA₂ and the toponyms at Haghia Triada; M157, the institution sign, on 278 tablets at Susa; the institution and office signs at Uruk) and close with a total on part of the tablets (KU-RO at Haghia Triada; the totals of Susa and Uruk marked by position). The proportion of entries with a quantity differs among the three (0.60, 0.75 and 0.80; $\chi^2 = 121$, $p < 10^{-26}$ on 8,282 and 23,272 entries), but the order of magnitude is the same and the difference is explained by the proportion of headings and broken lines in each corpus; what is comparable is the anatomy, not the rate.

3.1 bis Comparative table

instrument	Haghia Triada (224 docs)	Susa (1,594 tablets)	Uruk (6,387 tablets)
entries with a quantity	0.60	0.75	0.80
string singletons (1 sign / 3+ signs)	0.03 / 0.76	0.03 / 0.87-0.99	0.03 / 0.9
within-document recurrence with different quantity (short / long)	0.13 / 0.02	0.13 / 0.002	comparable to Susa
institutional header	SA-RA ₂ (20 docs), toponyms	M157 (278 tablets)	institution and office signs
total	KU-RO (37), PO-TO-KU-RO, DA-I	by position	by position
vocabulary monopolies	6 (accounts) + 3 (allocations)	24 (capacity), 3 (decimal)	pending
co-exclusion with curveball null	0 pairs (no power)	13 pairs	pending
frame of long strings	KA- 17% on roundels; -JA 7 pairs	M288 final (lift 3.8); M124 initial	pending
numeral systems by class	its own fractions (Corazza et al. 2021)	4 systems, own decimal	5 systems, source of Susa's
attributed hands	21 (73 tablets)	none in the catalogue	none

3.2 The partition of the lexicon is the same, and is seen better with more data

In the three corpora short strings are few, frequent and repeated, and long strings are many, unique and not repeated. At Susa the gradation is clean: one-sign strings are singletons in 3% of cases and recur within the tablet with different quantities in 13%; strings of three or more signs are singletons in 87-99% and recur in 0.2%. At Haghia Triada the same partition exists (commodity and transaction terms recur, names are singletons at 76%) but is seen with less definition because the short class is mostly logographic. Uruk behaves like Susa. The partition between what is counted and who delivers or receives it is therefore a feature of early bookkeeping and not of one tradition, and the behavioural criterion (recurrence with different quantities) recovers it without reading in all three.

3.3 The compartments, and who makes them

At Haghia Triada, the accounts with a total have six monopoly words and the allocations three ($p < 0.01$ in both cases); at Susa, the capacity system (grain) has 24 monopoly strings, the decimal 3, the sexagesimal none. Co-exclusion, which at Haghia Triada is not measurable (zero pairs with expected ≥ 1 in 224 documents), appears at Susa with 830 tablets: 13 pairs never co-occur when they should (the header M157 with M371, M370, M373, M046, M367; the grain product M297 with M376, M124, M003, M032, M009), which separates grain accounts from those of livestock and personnel. Only Haghia Triada allows the control by hand: with 73 tablets attributed to 21 scribes, shared vocabulary is explained by the hand (effect 0.032, $p = 0.002$ controlling for type) and not by the document type (0.025, $p = 0.21$): the Minoan compartments are, at least in part, scribal portfolios. Susa and Uruk have no attributed hands in the catalogue, and their compartmentation cannot be decomposed in this way; it is stated.

3.4 Long strings have a frame

At Susa, the 1,296 strings of three or more signs with a quantity have an institution or object sign in final position above chance (M288 231 times, lift 3.8, $p < 10^{-6}$; also M263, M297, M346) and a class sign in initial position (M124, M305, M157). The “names” carry a frame: classifier in front, variable core, institution or object behind. At Haghia Triada the frame exists with less power: the prefix KA- (17% of roundel groups against 4% of tablet groups) and the suffix -JA (seven root/derivative pairs against 3.1 expected). It is the same phenomenon at two scales of data.

3.5 The numerals

Susa inherits from Uruk the sexagesimal system (identical chain $N45 \rightarrow N34 \rightarrow N14 \rightarrow N01$, same carries), adapts the capacity system (same upper segment $N01 \rightarrow N39 \rightarrow N24$, its own fraction segment $N30C, N30D, N39C$), invents the decimal ($N23$ and the chain $N23 \rightarrow N14 \rightarrow N01$ do not exist at Uruk; $N51$ changes value, from 120 bisexagesimal to 1,000 decimal) and does not take the bisexagesimal (Uruk’s chain $N19 \rightarrow N04$, 231 cases, absent at Susa). It is the description of Damerow and Englund and of Dahl, recovered by structure, with the correction that the bisexagesimal inheritance is not confirmed. Linear A has no partner for comparison on this side: its fractions (Corazza et al. 2021) are a system of its own.

3.6 The periphery

At Susa the institutional header M157 is exclusive to the centre (278 tablets, 0 at Malyan, Yahya, Sialk or Sofalin); Yahya writes almost only grain accounts and has six signs of its own; Malyan uses the Susa inventory without a foreign sign. In Crete, Haghia Triada against Khania and Zakros shows a difference of repertoire of the same kind (the HT monopolies do not appear outside), with the same limitation of size of the periphery. In both cases the centre/periphery difference is one of administrative genre before one of script.

3.7 Note on the scope of this paper

The comparison of which administrative functions each archive puts into words (totals, transfer, sealing, date, measure), and its consequences for the question of whether the form of the Minoan receipt is native or borrowed, are treated in a separate paper (Acedo 2026g), which adds to these three archives the Middle Assyrian archival texts, the Amarna letters and the Neo-Assyrian records. The object here is different: which distributional instrument begins to see at which corpus size.

4. Discussion

What is common. The early bookkeeping of three regions without direct contact (Uruk and Susa had it; Crete with neither) has the same anatomy and the same partition of the lexicon, and in the three cases the part that records persons and places is the open class of long, unique strings, and the part that records what is counted is the closed class of short, repeated signs. That is not linguistics: it is the shape a record of deliveries takes when it is written, and it is seen in the three scripts because the three were born for that.

What is proper to each. Susa invents a decimal system for persons and animals and does not take the bisexagesimal; Crete writes the commodity almost always with a logogram and the party with syllables, and separates receipts (roundels) from lists (tablets) with a prefix; Uruk has the systems of time, rations and area that the other two do not need. Each archive shows its economy in what it decided to count and in how it decided to count it.

One negative worth recording. When the Susa tablets are ordered by the volume of commodity they record, vocabulary diversity does not change (singleton rate 0.40, 0.40, 0.36 and 0.42 by quartiles; $p = 0.47$): what grows with volume is repetition within the tablet (from 0.08 to 0.49), not lexical dominance. The analogy with the energy gradients of ecological systems, which predicts less diversity at higher input, does not transfer to the archive.

What it teaches about the instrument. Co-exclusion, the frame of names and the clean partition by length appear clearly at Susa and Uruk and not at Haghia Triada, whose size is a fifth and whose lexicon is mostly logographic. The control by hand, on the other hand, is possible only at Haghia Triada, because only there are scribes attributed. Each corpus answers the instruments that its size and its metadata allow, and the comparison fixes for the first time, with a number, the order of magnitude at which each instrument begins to see: co-exclusion, around eight hundred samples with mixed vocabulary; frame of names, around a thousand long strings; control by hand, with twenty hands and seventy documents.

An asymmetry that conditions all of the above. The three corpora do not offer the same metadata: only Haghia Triada has attributed hands, so that the scribal confounder can be measured there alone; only Susa and Uruk share a comparable numeral system, so that inheritance can be tested between them alone; and only Susa and Uruk have the size for co-exclusion. Every comparative claim in this paper holds for the corpora in which the corresponding instrument has data, and no further. What it does not say. Nothing about the languages, which remain three unknowns, nor about kinship between the scripts, which have none save Uruk-Susa. What it says is where each archive stands in the typology of early written administration, and that Linear A, the smallest of the three, is a normal archive of that typology, measured with the yardstick its elder siblings allow one to calibrate.

Data and reproducibility

Corpora, scripts and tables accompany the paper. Linear A: lineara.xyz (Hogan 2022) and SigLA; Proto-Elamite and proto-cuneiform: CDLI (cdli.earth), exports of September 2026 included.

Declaration of assistance

Corpus processing, permutation tests and the first draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

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